Lawrence Feng

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Education

Carnegie Mellon University **Pittsburgh, Pennsylvania**

M.S. in Machine Learning *Leave of Absence*

B.S. in Statistics and Machine Learning; Minor in Computer Science; GPA: 3.90 *May 2026*

- Selected coursework: Introduction to Deep Learning; Deep Reinforcement Learning; Probabilistic Graphical Models; Probability and Mathematical Statistics; Algorithm Design and Analysis; Advanced Methdos for Data Analysis.

Publications

Early Data Exposure Improves Robustness to Subsequent Fine-Tuning. **Lawrence Feng**, Gaurav R. Ghosal, Jacob Mitchell Springer, Ziqian Zhong, and Aditi Raghunathan. *Conference on Language Modeling (COLM)*, 2026.

Regularization Amplifies Subliminal Learning. **Lawrence Feng**, Jacob Mitchell Springer, Gaurav R. Ghosal, Ziqian Zhong, and Aditi Raghunathan. Under review at *International Conference on Learning Representations (ICLR)*, 2027.

Research Experience

Research Assistant **May 2026 – Nov. 2026**

Carnegie Mellon Machine Learning Department; advised by Professor Aditi Raghunathan

- Solo-authored the technical report "Subliminal Learning Happens at Every Rank, Given the Right Learning Rate and Enough Data", identifying and addressing methodological issues in prior work.
- Our follow-up paper shows a more general result. Researchers have found our work clarifying and surprising.

Honors Undergraduate Research Thesis **August 2025 – May 2026**

Advised by Professor Aditi Raghunathan

- Led the project and conducted all experiments for a first-author COLM 2026 paper.
- Reframed catastrophic forgetting as a property of upstream training rather than solely a downstream mitigation problem: *how* a model acquires a capability matters for how it is retained, not just whether it acquires it.
- Introduced *early exposure*—mixing target data into pretraining—and demonstrated a *latent* benefit: it mitigates forgetting during downstream fine-tuning even when it provides no measurable benefit before fine-tuning.
- Developed theory showing that early exposure protects specialized features from downstream interference.

Watermarking for Language Models **Dec. 2024 – June 2025**

Carnegie Mellon Computer Science Department; advised by Professor Wenting Zheng

- Developed a more efficient sampling algorithm for embedding provably quality-preserving watermarks into LLM outputs.
- Owned experimental design, implementation, analysis, and progress reporting as the project's sole programmer.
- Synthesized results into clear visualizations and technical reports for collaboration with non-ML researchers.

Robotics Institute at Carnegie Mellon **Spring 2024 – Fall 2024**

Summer Research Apprenticeship

- Explored adapting CoTracker's retrospective dense point tracking into predictive models that output future robot states.
- Uncovered significant robustness failures in state-of-the-art models through systematic environmental variations.

Industry Experience

Anthropic Fellows Program **November 2026 - Present**

AI Safety Fellow

Inflection AI **June 2025 - August 2025**

Member of Technical Staff Intern

- Implemented Qwen2.5-VL (PR #2868) within the open-source torchtune library, enabling distributed fine-tuning.
- Designed an automatic benchmark curation tool that increased the statistical power of LLM-as-a-judge evaluations of stylistic properties of LLM outputs; this tool was adopted into the company's core evaluation suite.
- Contributed to a company-wide emotional-intelligence evaluation effort by surveying prior work, improving benchmarks, prototyping persona-based user simulators, and building synthetic-data pipelines.

Leadership

Carnegie AI Safety Initiative **Fall 2023 – Nov. 2026**

President (August 2025 – May 2026), Member

- Spearheaded outreach growing mailing list 4× to ∼ 900, expanded leadership team 4× to ∼ 16, developed and taught technical AI safety fundamentals course to ∼ 40 ppl/sem, piloted and grew jailbreaking hackathon to ∼ 200 participants

Technical Skills

Programming Languages: Python, R, C, C++, Java, SQL, SML, LaTeX, Bash
Developer Tools: GCP, AWS EC2, Git, Weights & Biases, Slurm, Conda, uv
Frameworks and Libraries: PyTorch, TensorFlow, Hugging Face, vLLM, OpenCV, NumPy, Pandas, Matplotlib, ROS